

Supplier Performance Analytics & Risk Prediction System

Machine Learning-Based Decision Support System for Supply Chain Optimization

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Key Performance Snapshot

This dashboard provides end-to-end visibility into supplier delivery performance, powered by trained ML models — not simple statistics.

18.8

Average Lead Time

Days

7.1

Lead Time Std Dev

Days

7

Fastest Lead Time

Days

40

Slowest Lead Time

Days

Why Supplier Performance Monitoring Matters

Unpredictable supplier performance creates operational disruptions that cascade through the entire supply chain. Without proactive visibility, organizations remain reactive rather than strategic.

Unpredictable Lead Times

Fluctuating delivery windows impact production planning and inventory optimization

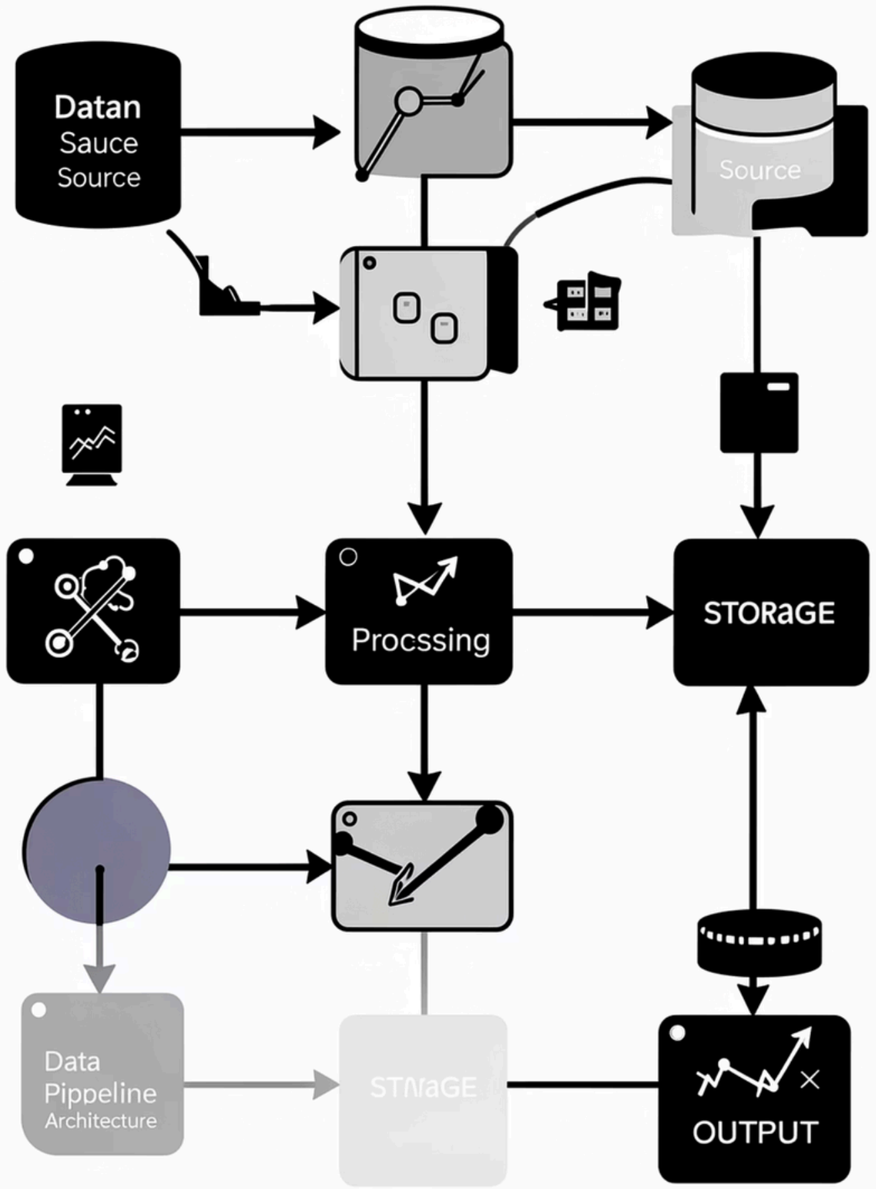
PO Delays Without Early Warning

Reactive instead of proactive management leads to costly disruptions

Lack of Supplier Risk Classification

Difficult to prioritize follow-up actions and allocate resources effectively

> A data-driven decision support system is needed to predict, monitor, and act — before disruption happens.



Data Sources & Dashboard Architecture

1

Data Sources

PO.csv + GR.csv + supplier_master.csv

2

Data Processing

Feature Engineering & Validation

3

ML Model Training

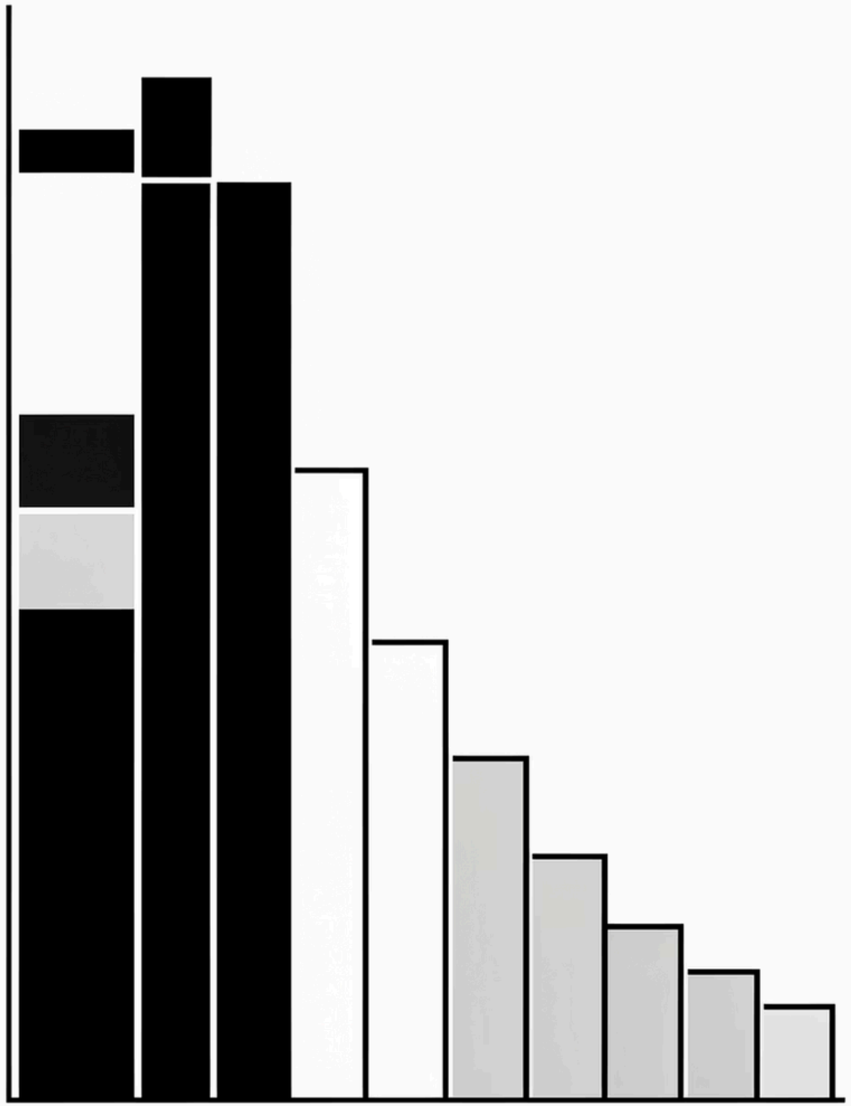
Lead Time Regression + IsLate Classification

4

Streamlit Dashboard

3 Interactive Pages

Models saved as .pkl files; features as .json/.pkl. Bilingual support (ID/EN).



Understanding Lead Time Patterns

Key Statistical Insights

- Distribution: Right-skewed, peak between 15–20 days
- High variability (Std Dev 7.1 days) signals unreliable supplier commitments
- Electronic category tends to have the longest lead times
- Category-based segmentation supports differentiated procurement strategies

Category Breakdown

Chemical, Packaging, Electronic show distinct LT profiles requiring tailored procurement approaches.

Operational Implication

The right-skewed distribution with high standard deviation indicates significant unpredictability in supplier commitments, necessitating buffer stocks and proactive monitoring.

Top 10 Suppliers with Slowest Lead Time

Supplier ID	Avg LT (days)	Std LT	Order Count
SUP024	27.86	2.83	213
SUP033	27.31	2.68	197
SUP014	26.47	2.44	209
SUP004	26.46	2.49	216
SUP034	26.40	2.08	199
SUP021	25.92	2.27	211
SUP013	25.84	2.15	205
SUP003	25.73	2.31	220
SUP012	25.68	2.09	208
SUP002	25.42	2.18	214

⚠ Suppliers with consistently high lead time AND high order count represent the highest operational risk and should be prioritized for contract renegotiation or alternative sourcing.

Predicting Lead Time Before PO is Placed

01

Input Features

Supplier ID, PO Date, Expected Delivery Date, Quantity

02

Feature Engineering

Supplier historical stats (Avg LT, Late Rate, Defect Rate, Reliability, Std LT), temporal features (Month, Quarter, Year), Category & Region encoding via Target Encoding

03

ML Prediction

Predicted Lead Time (days) + deviation from supplier promise

Example Scenario

- Supplier: SUP001
- PO Date: 2026/05/04
- Expected Delivery: 2026/05/18

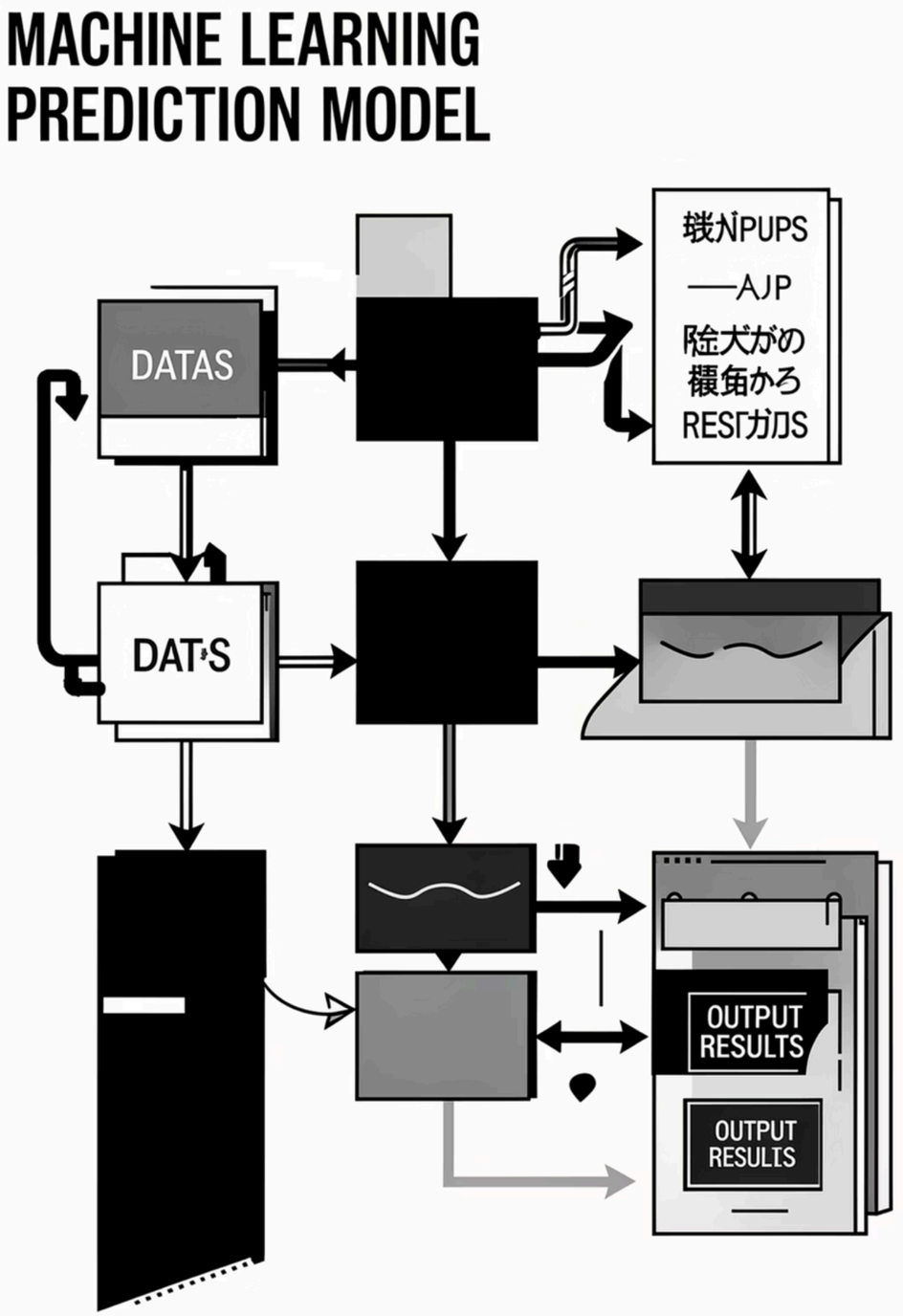
Prediction Result

Predicted Lead Time: 22.0 days

Supplier Promise: 14 days

Deviation: 8 days SLOWER than promised

Historical Average: 22.0 days (consistent with prediction)



Early Warning System for PO Delay Risk

The IsLate binary classifier provides probability-based risk tiers with actionable recommendations.

🚨 HIGH RISK (prob > 70%)

Consider alternative supplier or add safety stock. Immediate action required.

⚠️ MEDIUM RISK (40–70%)

Proactive follow-up with supplier recommended. Monitor closely.

✅ LOW RISK (< 40%)

Proceed with PO as planned. Standard monitoring sufficient.

Dual-Validation Logic

Model prediction aligned with rule-based logic

Predicted LT vs promised LT → High confidence

Conflict between model and rule

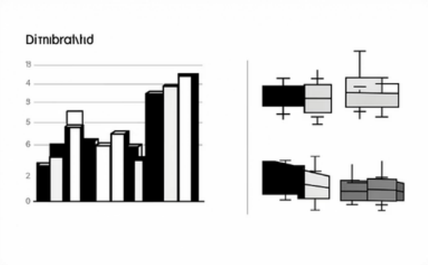
Flag for manual audit and supplier communication

3-Page Interactive Dashboard Built with Streamlit



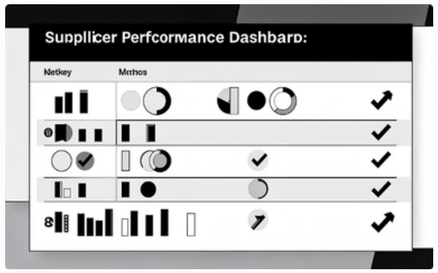
Main Dashboard

- KPI cards: Avg LT, Late Rate, OTIF, Total Suppliers
- Lead Time vs Late Rate trend chart (monthly)
- Supplier performance scatter: LT vs Late Rate
- Top 5 late suppliers



Lead Time Analysis

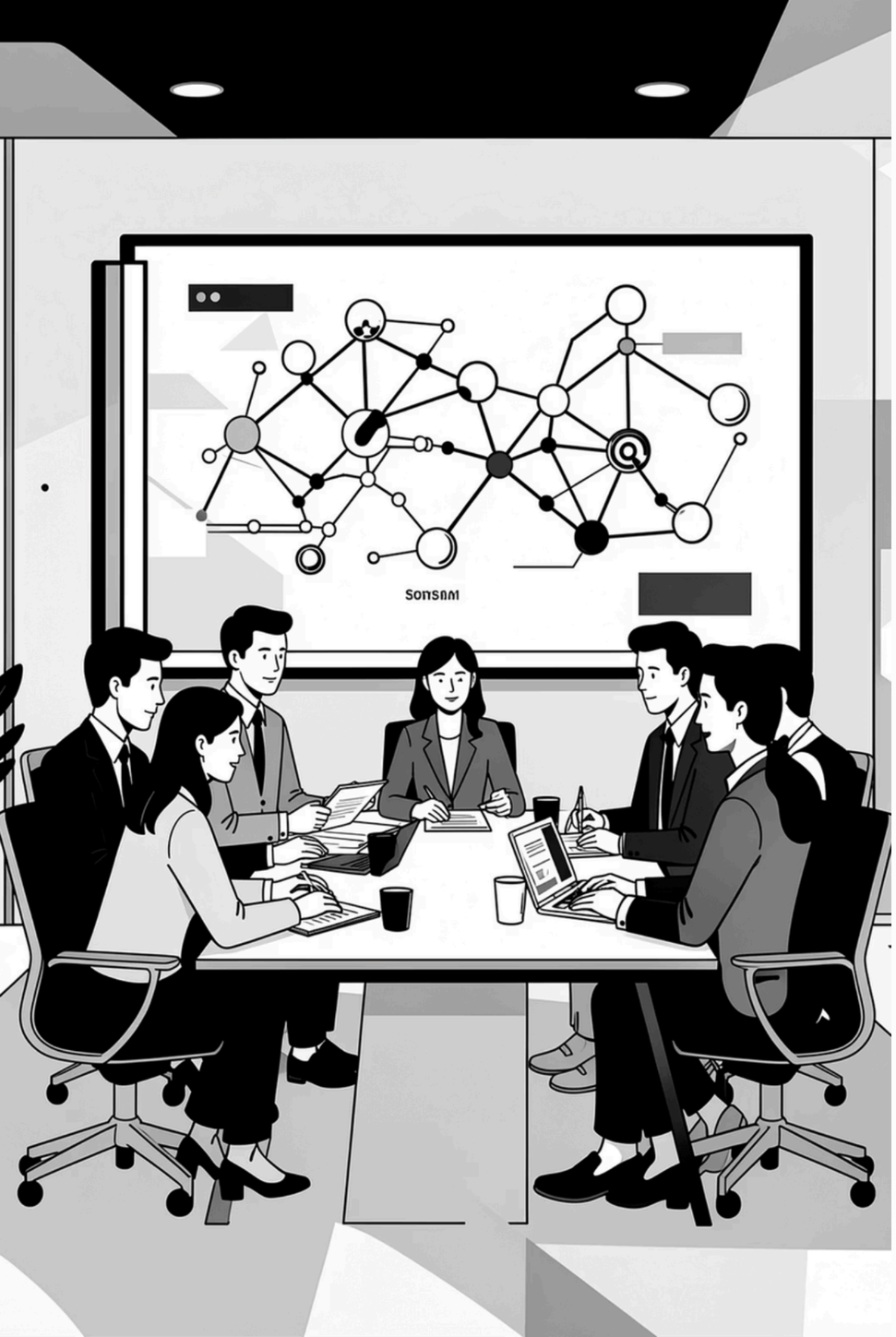
- Distribution histogram
- Lead Time by Category (boxplot)
- Lead Time by Supplier (bar chart)
- Predict Lead Time (ML-powered form)



Supplier Analysis & PO Prediction

- Supplier performance summary table
- PO delay risk prediction form
- Risk recommendation output

Additional Features: Bilingual UI (Bahasa Indonesia / English), dark theme, fully interactive



From Insight to Action

What Was Achieved

- End-to-end supplier monitoring system with ML prediction
- Dual-model approach: Lead Time Regression + Delay Classification
- Interactive DSS for procurement decision-making
- Supplier risk tiering with actionable recommendations

Recommended Next Steps

1. Integrate real-time PO data pipeline (e.g., ERP connection)
2. Add defect rate monitoring as third ML model output
3. Implement supplier scorecard with composite KPI (LT + OTIF + Defect Rate)
4. Deploy to cloud (Streamlit Cloud / internal server) for team-wide access
5. Periodic model retraining as new PO/GR data accumulates

> *"Data-driven procurement starts with knowing your suppliers — before they surprise you."*

Open Dashboard